

Received 9 May 2025, accepted 30 May 2025, date of publication 4 June 2025, date of current version 12 June 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3576086

RESEARCH ARTICLE

Gradient Boosting-Based Simultaneous Classification and Regression Approach

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ABSTRACT The recent development of advanced data analytics and machine learning promoted the introduction of diverse learning techniques designed to alleviate challenges related to two major supervised learning tasks: (i) Classification and (ii) Regression. Despite the successful consideration of these tasks to address problems relevant to various applications, the design of dual machine learning approaches that perform both tasks simultaneously remains challenging. This paper proposes a Dual-learning XGBoost Based Approach (DXGBA), a novel gradient boosting-based machine learning approach that concurrently carries out classification and regression tasks. Specifically, the proposed approach aims at learning a supervised learning model that is jointly trained for both classification and regression using a single unified framework. The optimization of this dual task is formulated using a novel joint cost function that minimizes the total error of both classification and regression tasks during the learning phase. The proposed approach was assessed using benchmark datasets and performance measures. The experiments proved its effectiveness compared to native eXtreme Gradient Boosting (XGBoost) models. DXGBA achieved an F1-score of 0.4718 and an MSE of 0.0002 on the bankruptcy dataset, outperforming the single-task XGBoost, which yielded 0.3942 and 0.0003, respectively. Moreover, DXGBA reduced testing time by up to 81.9% compared to baseline methods. It also evidenced better performance relative to conventional machine learning models.

INDEX TERMS Classification, dual machine learning, dual XGBoost approach (DXGBA), gradient boosting, regression, supervised learning, XGBoost.

I. INTRODUCTION

In recent years, data science has emerged as an essential discipline for uncovering patterns, extracting knowledge, and building predictive models from vast and complex datasets [1]. Technological advancements have driven a rapid development in data generation [2], [3], with global data volumes increasing by approximately 50% annually [3]. This growth poses significant challenges in data storage, processing, and analysis, often beyond the capabilities of traditional statistical methods [4], [5], [6]. Consequently, machine learning (ML) and artificial intelligence (AI) have emerged as powerful alternatives [7]. ML, in particular, has demonstrated exceptional efficiency in sectors such as healthcare, finance, and manufacturing, enabling automated decision making, operational optimization, and service personalization.

The associate editor coordinating the review of this manuscript and approving it for publication was Mostafa M. Fouda^{ID}.

Within supervised learning, classification and regression are two core tasks focused on predicting categorical and continuous outputs, respectively [7]. For supervised machine learning, the researchers' efforts have been prototypically intensified to address either classification or regression tasks. While both have been traditionally addressed using distinct models and evaluation methods, real-world applications increasingly demand more integrated and sophisticated solutions that require both classification and regression to be performed simultaneously. For example, in the domain of prosthetics, machine learning algorithms are used to determine both the type of finger movement and estimate the applied force [8]. In medical diagnostics, it is often essential to determine whether a patient has a disease while also predicting a clinical severity score [9]. Similarly, in financial applications such as loan approval or bankruptcy prediction, it is necessary to evaluate the risk status (e.g., bankrupt or not) in addition to predicting a

quantitative value (e.g., financial priority score or approved loan amount) [10].

Recent advances in machine learning have led to highly optimized algorithms for both classification and regression tasks. However, a critical limitation persists in that most existing models, including powerful frameworks like eXtreme Gradient Boosting [11] (XGBoost), treat these two tasks as mutually exclusive. Specifically, conventional supervised learning approaches typically train separate models for classification and regression, resulting in isolated learning processes that fail to exploit potential synergies between the two. This siloed approach leads to inefficiencies, especially in domains where both a categorical and a continuous outcome are derived from the same input and are inherently interrelated. Despite the widespread adoption of Gradient Boosting Machines [12] (GBMs) for their robustness and accuracy, and the proven superiority of XGBoost in many practical applications, these models lack native support for joint task learning. The absence of a unified framework capable of handling classification and regression simultaneously presents a substantial gap, particularly in contexts where shared feature representations can improve predictive accuracy and resource efficiency. To bridge this gap, we propose DXGBA, a Dual-learning XGBoost Based Approach that extends the classical XGBoost to simultaneously optimize both tasks through an integrated objective function, effectively leveraging shared patterns to improve overall performance.

The primary contributions of this study are summarized as follows:

- Proposing DXGBA, a novel dual-learning gradient boosting framework that extends XGBoost to simultaneously perform regression and classification using a shared decision tree structure.
- Developing a joint custom objective function that integrates classification and regression loss components to leverage inter-task dependencies and guide the model toward mutually beneficial feature representations.
- Evaluating the effectiveness of DXGBA by comparing it against traditional single-task learning models and state-of-the-art methods using multiple benchmark datasets from diverse application domains.
- Analyzing computational efficiency, including both predictive performance and testing time, to demonstrate the practical advantages of the proposed approach.

The rest of this paper is organized as follows: In Section II, the existing conventional and deep learning approaches that automatically perform simultaneous classification and regression are surveyed. In Section III, the proposed approach, encompassing its loss formulation, optimization methodology, and implementation details, is described. Finally, experimental design, datasets, assessment criteria, and comparative findings are presented in Section IV.

II. LITERATURE REVIEW

Numerous endeavors have been made to integrate joint classification and regression within the realms of both

traditional machine learning and contemporary deep learning paradigms. While some researchers have explored the concurrent output of classification and regression values, their models often adopt an independent learning approach, relying on two distinct algorithms—one for classification and another for regression. Moreover, a predominant focus in existing research has been on utilizing the output of one model to enhance the performance of the other. To the best of our knowledge, no prior studies have systematically addressed the challenge of unified learning using conventional machine learning methodologies, particularly within the context of boosting techniques.

A. CONVENTIONAL APPROACHES

The authors in [8] addressed the problem of prosthetic hand control by deriving a model that predicts finger motion and force simultaneously from the forearm surface electromyogram (EMG). However, the proposed model is composed of two sub-models for finger motion classification and force regression that are separately trained. First, the sEMG feature vector is passed to an Artificial Neural Network (ANN) to classify finger motion. Then, for finger force estimation, both EMG signals and force signals are used to build linear and quadratic models, and the best-fitted model is selected based on the Mean Absolute Error (MAE) between the actual force and the predicted one.

The study in [13] proposed a classification-regression architecture that models the multi-stage Remaining Useful Life (RUL) prediction of a system. Given multiple types of raw sensory data, features are extracted and then passed to two modules: a classification module and a regression module. Support Vector Machine (SVM) and Support Vector Regression (SVR) algorithms are adopted for the classification and regression modules, respectively. The classification module is used to identify the current health stage of the system and classify its status into one of three classes: new, medium, or heavy. Alternatively, the objective of the regression module is to predict the value of the stage-level remaining useful life. The SVR model is trained using the output from the health stage classification module along with other predictor variables. The feature vector is then passed to the SVR model to predict the failure time of the current stage.

In [14], an approach for unilateral transradial amputees to predict the wrist force from surface electromyogram is introduced. Using a Multiple Kernel Learning (MKL) algorithm, feature vectors extracted from the EMG signals are first classified into three classes: positive output, negative output, and dead zone output. Then, an ANN is trained for each class as a regression model to predict the corresponding value of the wrist force. The objective of this study is to enhance the predictive performance of force estimation by incorporating a classification step before the regression process.

In the study outlined in [15], a framework was introduced for finger force regression, incorporating finger movement classification as a guiding step using a Mixture of Experts

(MoE) regression model. Specifically, the classification of EMG signals is performed through the utilization of Gaussian Mixture Models (GMMs) as gates. Each GMM is linked to a specific finger movement, selecting the gate with the highest a posteriori probability for generating the feature vector. Subsequently, based on the chosen gate, the feature vector is allocated to the corresponding expert for finger force regression.

B. DEEP LEARNING-BASED APPROACHES

The study in [10] proposed a deep learning approach that addresses the problem of determining companies expected to face bankruptcy and prioritizing the financial support to be offered by governments or financial institutions where their financial performance is more critical. More specifically, based on a deep neural network, the proposed model classifies companies into bankruptcy and non-bankruptcy classes and predicts the expected financial performance of the companies (e.g., total sales). The deep neural network is composed of an input layer, a set of hidden layers, and two output layers, one for classification and another for regression. For a single input, the model predicts both a class label (bankruptcy or non-bankruptcy) and a numeric value indicating the performance of the company via the output layers. The model is trained to optimize a joint objective function.

The authors in [9] also introduced a deep learning framework based on Convolutional Neural Network (CNN) for joint classification and regression. Using Magnetic Resonance (MR) imaging and demographic information of subjects as the input data, the model performs two types of tasks: multi-class brain disease classification and regression of four clinical scores. The developed architecture is composed of a set of convolutional layers followed by two Fully Connected (FC) layers. The output of the last FC layer is then concatenated with the demographic information and fed into two FC layers. Lastly, the output of the last FC layer is propagated to two parallel FC layers to predict the class label and estimate the values of the clinical scores, respectively.

The authors in [16] designed a deep learning framework to perform simultaneous hand gesture classification and finger angle estimation based on a single model. More specifically, the proposed model detects hand gestures and identifies which finger is flexing during finger movement, along with estimating the joint angle of each finger. A dual-output CNN model is employed where the last layer is composed of two parallel dense layers. The first uses K neurons for hand gesture classification, and the second uses a single neuron for finger angle regression, where K is the number of hand gesture classes. Upon that, the output of the classification dense layer is propagated to a softmax activation function to obtain class probabilities. A weighted combination of the regression loss and classification loss is used to train the model.

Furthermore, the work in [17] addressed the problem of online action detection from streaming 3D skeleton data.

A deep learning end-to-end model is built to integrate both classification and regression, where, in addition to identifying the action label, the model also predicts the starting and ending time of the action from the streaming data. The proposed model is composed of a shared deep Long-Short Term Memory (LSTM) network, a classification subnetwork, and a regression subnetwork. The classification subnetwork is first trained for action classification, then, in addition to the classification output and under its guidance, the regression subnetwork is trained to attain more accurate estimation of start and end time.

The study in [18] also designed a multi-task learning architecture to address the problem of online handwriting recognition. A Multivariate Time Series (MTS) is classified into a character class, and the handwriting trajectory is reconstructed to be as properly aligned as possible with its ground truth. The multivariate time series is passed to a CNN for feature extraction. Features are then processed by two different task-specific separate layers (heads) composed of an LSTM and dense layers for each task. The output of the model is the predicted class label of the MTS along with the predicted MTS.

The authors in [19] presented a joint learning approach for age prediction of online users from textual and social features. The classification task aims to classify users into different age groups, whereas the regression task focuses on predicting a discrete variable that represents the exact age. In detail, the proposed algorithm is composed of a main LSTM layer and an auxiliary LSTM layer, which is shared by both the main and the auxiliary tasks. Features are extracted and fed into both the main and the auxiliary LSTM layers, which are followed by a fully connected layer. The output of the auxiliary LSTM layer is then fed into a softmax layer for probability prediction and classification. For age regression, the output obtained from the auxiliary LSTM layer is concatenated with the output of the main LSTM layer to form the regression representation.

As one can see, researchers have investigated a variety of approaches to the problem of performing simultaneous regression and classification. Generally, deep learning-based approaches rely on the idea of utilizing two parallel dense layers as outputs. One criticism of much of the literature is that classification and regression models are trained independently of each other, although it is reported to be simultaneous. For example, the proposed architectures in [13], [14], and [15] are composed of two different models separately trained, with one model exploiting the output from the other model. Furthermore, the approach introduced by [17] trains the classification model first and then fine-tunes the entire model over the initialized classification model. The approach proposed by [19] takes the advantages of both classification and regression models through employing an auxiliary classification model to improve the performance of the main regression model by learning more accurate features. In [8], the framework is also designed to have two models learned individually with different feature vectors

TABLE 1. Summary of the reviewed approaches.

Ref	Approach	Models Used		Input Data	Application-based	Learning Process	Task Exploiting the Other
		Classification	Regression				
[8]	Conventional	AN	Linear Regression	Raw data	Yes	Separate	No
[13]	Conventional	SVM	SVR	Raw sensory data	Yes	Separate	Yes
[14]	Conventional	MKL	ANN	EMG signals	Yes	Separate	Yes
[20]	Conventional	SVM	MLR	Zero-inflated time series	No	Separate	Yes
[15]	Conventional	MoE	GMM	EMG signals	No	Separate	Yes
[9]	Deep Learning	CNN	CNN	MR image patches	Yes	Joint	No
[19]	Deep Learning	LSTM	LSTM	Textual and social features	No	Joint	Yes
[16]	Deep Learning	CNN	CNN	Images	Yes	Joint	No
[17]	Deep Learning	LSTM	LSTM	Videos	Yes	Separate	Yes
[18]	Deep Learning	CNN	CNN	Multivariate time series	Yes	Joint	No
[10]	Deep Learning	Deep NN	Deep NN	One-dimensional data	No	Joint	No

for each task. The authors in [20] demonstrated that the classification and regression models are trained to optimize a joint objective function. Nevertheless, the objective function is minimized to learn the parameters of the regression model, and then, upon convergence, the classifier is trained with the input features along with the predicted value from the regression model to output a class label. In fact, this study would have been more relevant if both models were jointly trained.

Another limitation is imposed by some research where the methods are developed in such a way that the outcome of the regression and classification tasks is correlated. In particular, the authors in [21] presented an approach that performs simultaneous classification and regression. Specifically, a random forest-based model classifies anti-cancer drugs as effective or ineffective and quantifies the value of drug effectiveness as a response variable. As one can notice, both outcomes are relevant to each other as the regression output is the confidence score of the classification output. Table 1 summarizes the relevant studies that exploited conventional machine learning and deep learning approaches.

C. HYBRID BOOSTING WITH METAHEURISTICS

Recently, an increasing number of studies have shed light on hybrid boosting algorithms that employ metaheuristic optimization techniques to optimize the model's hyperparameters. This has resulted in enhanced performance and model interpretability across several application domains. For instance, the research in [22] introduced a two-level approach aimed at categorizing respiratory disorders through audio data. The first level is composed of a CNN model that extracts features and performs initial classification. In the second level, they replace the dense layer of the CNN with boosting algorithms, including XGBoost and AdaBoost [23] to perform the final classification. In both levels, the parameters of the model components are tuned using a modified Particle Swarm Optimization [24] (PSO) strategy, referred to as Genetic Diversity-Guided PSO (CDGPSO). In this strategy, Quasi-Reflexive Learning (QRL) initialization is utilized to ensure exploration of diverse solutions. Crossover and Mutation, inspired by Genetic Algorithms [25] (GAs), are

also incorporated to slightly perturb solutions. Furthermore, they dynamically update a threshold to balance exploration and exploitation.

The paper in [26] addressed vulnerabilities in IoT systems, particularly concerning the MQTT protocol. The motivation of this study is to develop an efficient detection model that can effectively generalize across different IoT-based threats. The methodology consists of three main components: feature reduction, intrusion detection using XGBoost, and hyperparameter optimization. They first reduce the dimensionality of the input features using a CNN model. Then, they employ XGBoost for traffic classification through binary and multi-class classification methods. Furthermore, they utilize a genetically inspired Reptile Search Algorithm (GIRSA), a modified version of the Reptile Search Algorithm [27] (RSA), to enhance the hyperparameters of the model. The algorithm is updated by using crossover and mutation to improve convergence rates and enhance diversity. Furthermore, they also apply a replacement method that substitutes the worst-performing individuals with newly developed solutions.

The methodology proposed by the paper in [28] introduced a Diversity Oriented Firefly Algorithm (DOFA), an improved version of the standard Firefly Algorithm [29] (FA), in a hybrid optimization framework to enhance phishing website detection. They use XGBoost as the base classifier. Additionally, they employ DOFA for feature selection by employing a binary version and hyperparameter optimization by employing the standard continuous DOFA. One of the most significant improvements to the algorithm is the utilization of QRL to guarantee diversity. Furthermore, the L1-norm metric is implemented to evaluate population diversity. Furthermore, they implement a dynamic reinitialization controlled by an adaptive threshold to replace the worst individuals. The threshold is reduced during iterations to balance exploration and exploitation.

The study in [30] also provided a comprehensive methodology that utilized sophisticated machine learning techniques to examine and analyze the atmospheric behavior of toluene. An XGBoost model is employed to predict toluene concentrations. Furthermore, they develop a hybrid metaheuristic

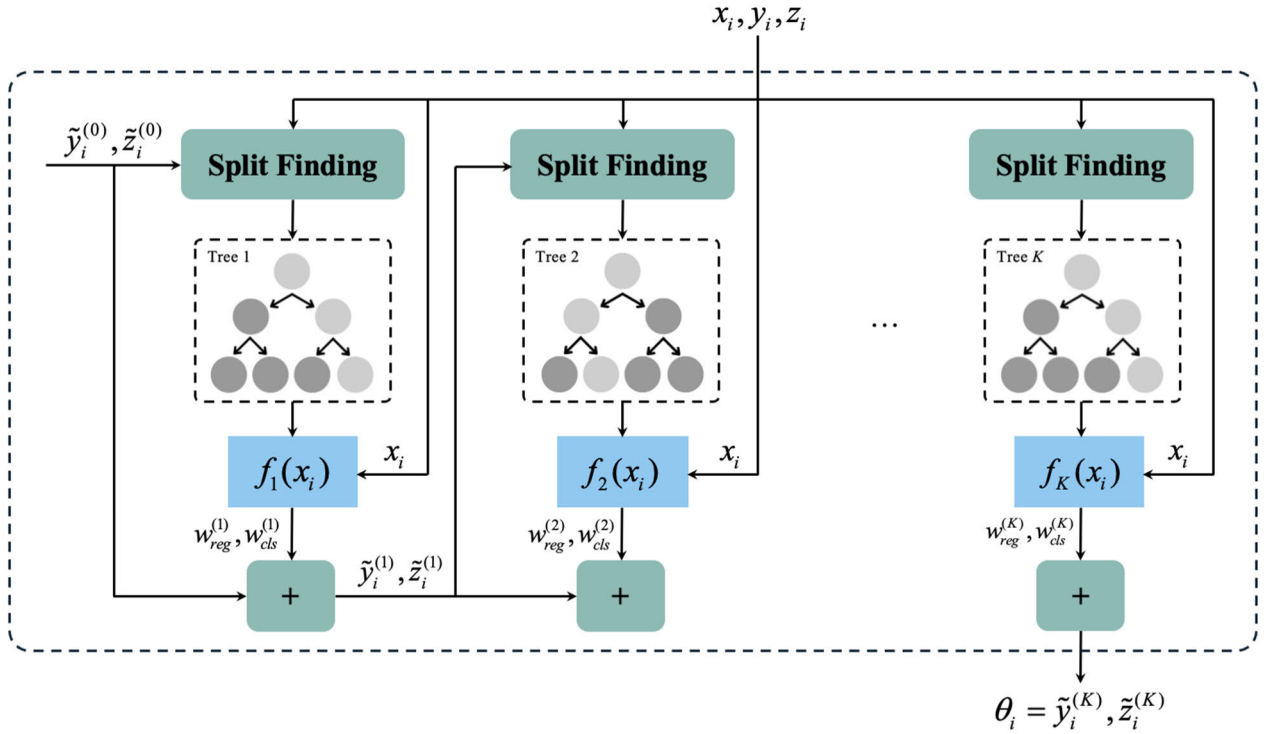


FIGURE 1. The block diagram of the proposed dual learning approach.

technique to optimize the hyperparameters of the model. More specifically, RSA [27] is combined with FA [29] to promote diversity and balance between exploration and exploitation. In addition, SHapley Additive exPlanations [31] (SHAP) is used on the best-performing XGBoost model to interpret model predictions and identify significant features. Similarly, the authors in [32] develop an XGBoost model that predicts external audit opinions using financial and non-financial data. They optimize performance using a metaheuristic to tune the hyperparameters. In particular, a boosted variant of the Sine Cosine Algorithm [33] (SCA) is developed, and SHAP is used to interpret model outputs.

III. PROPOSED APPROACH

In this study, a gradient boosting approach that simultaneously performs classification and regression is designed. It consists of three phases: (i) Building the dual model, (ii) Calculating the optimal weights, and (iii) Finding the best split. The block diagram of the proposed approach is depicted in Figure 1.

A. PROPOSED MODEL

Let $X = \{x_i\}_{i=1}^N$ be a set of N instances passed as input to the model, where x_i is an M -dimensional vector ($x_i \in \mathbb{R}^M$). Also, let y_i denotes the corresponding value of the target variable, where $y_i \in \mathbb{R}$ and $z^c = \{z_i^c\}_{i=1}^N$ ($c = 1, 2, \dots, C$) denote the labels of C categories.

Given an unlabeled sequence of instances, the proposed model aims to learn a nonlinear mapping $\psi : X \rightarrow (\mathbb{R}, \{z^c\}_{c=1}^C)$ from the input space to both the space of the value of the target variable and the class label. More specifically, to predict the output, the model uses K additive functions f_k such that:

$$\theta_i = \psi(x_i) = \theta_i^{(0)} + \sum_{k=1}^K \alpha f_k(x_i), f_k \in F \quad (1)$$

In (1), $\theta_i = \{\tilde{y}_i, \tilde{z}_i\}$ is the output of the model that composed of the predicted value of the target variable and the predicted class label, respectively. On the other hand, $\theta_i = \{\tilde{y}_i^{(0)}, \tilde{z}_i^{(0)}\}$ represents the initial default values. Given J leaves in the tree, $F = \{f(x) = w_{q(x)}\}$ indicates the space of the classification and regression trees, where $q: \mathbb{R}^M \rightarrow J$ and $w \in \mathbb{R}^J$. Finally, α is the learning rate.

The function q represents the tree structure that maps an instance to the corresponding leaf index. Each function f_k corresponds to a tree structure q and leaf weights w . In the proposed model, each leaf node j ($j = 1, 2, \dots, J$) has two continuous scores, referred to as weights: a classification weight w_{clsj} and a regression weight w_{regj} , such that $w_j = \{w_{clsj}, w_{regj}\}$. Thus, for a given instance, the decision rules within the trees, denoted by q , are employed to identify the relevant leaves and their associated weights, denoted by w . These weights are then summed up to predict the output.

Given an unlabeled sequence of observations, the goal is to learn a series of functions used in the model that best classify

each observation and estimate the value of its target variable by minimizing the following joint objective function L :

$$L(\psi) = \sum_{i=1}^N l_{reg}(y_i, \tilde{y}_i) + \frac{1}{2} \lambda w_{reg}^2 + \sum_{i=1}^N l_{cls}(z_i, \tilde{z}_i) + \frac{1}{2} \lambda w_{cls}^2 + \gamma J \quad (2)$$

where l_{reg} and l_{cls} are the regression loss and the classification loss, respectively, defined as:

$$l_{reg}(y_i, \tilde{y}_i) = \frac{1}{2} (y_i - \tilde{y}_i)^2 \quad (3)$$

$$l_{cls}(z_i, \tilde{z}_i) = -[z_i \log(\tilde{z}_i) + (1 - z_i) \log(1 - \tilde{z}_i)] \quad (4)$$

and λ is the regularization term. In (3), l_{reg} is the Mean Squared Error loss function that represents the performance of the regression task by calculating the difference between the target value y and the predicted value $\tilde{y}$. On the other hand, l_{cls} in (4) is the Cross-entropy loss that encloses the performance of the classification task. The second and fourth terms are regularization terms that prevent overfitting through weight smoothing. The last term, γJ , relies on γ , which is a user-defined penalty intended to stimulate tree pruning. One should mention that this term will be removed from the objective function since pruning occurs after building the tree, and hence it plays no role in deriving the optimal weight values or calculating similarity scores, which will be illustrated in the following subsections.

The model in (2) cannot be optimized by conventional optimization approaches since it includes functions as parameters. Instead, the model is trained iteratively by greedily adding w_{cls} and w_{reg} that most improve the model. Formally, let $\tilde{y}_i^{(t)}$ and $\tilde{z}_i^{(t)}$ be the predicted value and the predicted class of the i^{th} instance at iteration t , respectively. Thus, w_{cls} and w_{reg} of the t^{th} iteration needs to be added to minimize the following objective function $L^{(t)}$:

$$L^{(t)} = \sum_{i=1}^N [l_{reg}(y_i, \tilde{y}_i^{(t-1)} + w_{reg})] + \frac{1}{2} \lambda w_{reg}^2 + \sum_{i=1}^N [l_{cls}(z_i, \tilde{z}_i^{(t-1)} + w_{cls})] + \frac{1}{2} \lambda w_{cls}^2 + \gamma T \quad (5)$$

To promptly optimize the objective function in (5), a Second-order Taylor approximation is used:

$$\begin{aligned} l_{reg}(y_i, \tilde{y}_i^{(t-1)} + w_{reg}) &\approx l_{reg}(y_i, \tilde{y}_i^{(t-1)}) \\ &+ [\frac{\partial}{\partial \tilde{y}_i} l_{reg}(y_i, \tilde{y}_i^{(t-1)})] w_{reg} \\ &+ \frac{1}{2} [\frac{\partial^2}{\partial^2 \tilde{y}_i} l_{reg}(y_i, \tilde{y}_i^{(t-1)})] w_{reg}^2 \end{aligned} \quad (6)$$

$$\begin{aligned} l_{cls}(z_i, \tilde{z}_i^{(t-1)} + w_{cls}) &\approx l_{cls}(z_i, \tilde{z}_i^{(t-1)}) \\ &+ [\frac{\partial}{\partial \tilde{z}_i} l_{cls}(z_i, \tilde{z}_i^{(t-1)})] w_{cls} \\ &+ \frac{1}{2} [\frac{\partial^2}{\partial^2 \tilde{z}_i} l_{cls}(z_i, \tilde{z}_i^{(t-1)})] w_{cls}^2 \end{aligned} \quad (7)$$

Substituting (6) and (7) into (5) yields:

$$\begin{aligned} L^{(t)} &\approx \sum_{i=1}^N [l_{reg}(y_i, \tilde{y}_i^{(t-1)}) + g_{reg_i} w_{reg} + \frac{1}{2} h_{reg_i} w_{reg}^2] \\ &+ \frac{1}{2} \lambda w_{reg}^2 \\ &+ \sum_{i=1}^N [l_{cls}(z_i, \tilde{z}_i^{(t-1)}) + g_{cls_i} w_{cls} + \frac{1}{2} h_{cls_i} w_{cls}^2] \\ &+ \frac{1}{2} \lambda w_{cls}^2 \end{aligned} \quad (8)$$

where g_{reg_i} and g_{cls_i} are the gradients on the regression and classification loss function, defined as:

$$g_{reg_i} = \frac{\partial}{\partial \tilde{y}_i} l_{reg}(y_i, \tilde{y}_i) = -(y_i - \tilde{y}_i) \quad (9)$$

$$g_{cls_i} = \frac{\partial}{\partial \tilde{z}_i} l_{cls}(z_i, \tilde{z}_i) = -(z_i - \tilde{z}_i) \quad (10)$$

whereas h_{reg_i} and h_{cls_i} are the Hessians on the regression and classification loss function, defined as:

$$h_{reg_i} = \frac{\partial^2}{\partial^2 \tilde{y}_i} l_{reg}(y_i, \tilde{y}_i) = 1 \quad (11)$$

$$h_{cls_i} = \frac{\partial^2}{\partial^2 \tilde{z}_i} l_{cls}(z_i, \tilde{z}_i) = \tilde{z}_i(1 - \tilde{z}_i) \quad (12)$$

To simplify the objective function in (8), the constant terms are omitted, and the simplified objective function $\tilde{L}^{(t)}$ is expressed as follows:

$$\begin{aligned} \tilde{L}^{(t)} &= w_{reg} \sum_{i=1}^N (g_{reg_i}) + \frac{1}{2} w_{reg}^2 [\sum_{i=1}^N (h_{reg_i}) + \lambda] \\ &+ w_{cls} \sum_{i=1}^N (g_{cls_i}) + \frac{1}{2} w_{cls}^2 [\sum_{i=1}^N (h_{cls_i}) + \lambda] \end{aligned} \quad (13)$$

B. OPTIMAL WEIGHTS LEARNING

As outlined above, each leaf node in the tree contains continuous scores for regression and classification called weights. For a given instance, the corresponding weights are summed up to calculate the final prediction. To determine the value of the regression weight w_{reg} and classification weight w_{cls} , partial derivatives of the loss function with respect to both w_{reg} and w_{cls} are calculated. In particular, the partial derivative of the loss function with respect to w_{reg} is computed and set to zero. The derivative is given by:

$$\frac{\partial \tilde{L}}{\partial w_{reg}} = \sum_{i=1}^N (g_{reg_i}) + w_{reg} \sum_{i=1}^N (h_{reg_i}) + \lambda \quad (14)$$

Assigning the derivative in (14) to zero yields:

$$w_{reg} = \frac{-\sum_{i=1}^N (g_{reg_i})}{\sum_{i=1}^N (h_{reg_i}) + \lambda} \quad (15)$$

The equation in (15) yields the optimal regression weight w_{reg_j} for leaf j as:

$$w_{reg_j} = \frac{-\sum_{i \in I_j} (g_{reg_i})}{\sum_{i \in I_j} (h_{reg_i}) + \lambda} \quad (16)$$

where $I_j = \{x_i | q(x_i) = j\}$ is the set of instances in leaf j , λ is the regularization term, g_{reg_i} and h_{reg_i} are as defined in (9) and (11), respectively. As one can notice in (16), the regression weight is the average of the regression residuals in the leaf. Similarly, to calculate the optimal value of the classification weight, the derivative of the loss function with respect to w_{cls} is calculated, and assigning it to zero yields:

$$w_{cls} = \frac{-\sum_{i=1}^N (g_{cls_i})}{\sum_{i=1}^N (h_{cls_i}) + \lambda} \quad (17)$$

Thus, the equation in (17) yields the optimal classification weight w_{cls_j} for leaf j :

$$w_{cls_j} = \frac{-\sum_{i \in I_j} (g_{cls_i})}{\sum_{i \in I_j} (h_{cls_i}) + \lambda} \quad (18)$$

where $I_j = \{x_i | q(x_i) = j\}$ is the set of instances in leaf j , λ is the regularization term, g_{cls_i} and h_{cls_i} are as defined in (10) and (12), respectively. Algorithm 1 illustrates the steps of the model building and calculating the optimal weights.

Algorithm 1 Model Building Algorithm

Input:

$\{(x_i, y_i, z_i)\}_{i=1}^N$: training instances
 α : learning rate
 K : number of boosting rounds

Output:

ψ : trained model
 Initialize class probabilities: $\tilde{z}_i^{(0)} \leftarrow 0.5$
 Initialize regression predictions: $\tilde{y}_i^{(0)} \leftarrow \text{mean}(\{y_i\})$
 Set iteration counter: $k \leftarrow 1$

repeat

1. Compute regression residual: $r_{reg_i} \leftarrow y_i - \tilde{y}_i^{(k-1)}$
2. Compute classification residual: $r_{cls_i} \leftarrow z_i - \tilde{z}_i^{(k-1)}$
3. Create a new tree node j_k using residuals.
4. Find the best split using the Split Finding Algorithm 2.
5. Compute weights for regression and classification tasks for each leaf of the tree using (16) and (18).
6. Update regression prediction: $\tilde{y}_i^{(k)} \leftarrow \tilde{y}_i^{(k-1)} + \alpha w_{reg}^{(k)}$
7. Update classification prediction: $\tilde{z}_i^{(k)} \leftarrow \tilde{z}_i^{(k-1)} + \alpha w_{cls}^{(k)}$
8. Add tree j_k to the model ψ
9. Update $k \leftarrow k + 1$

until $k > K$

return ψ

C. BEST SPLIT FINDING

To build a tree in XGBoost, a greedy algorithm starts from a single leaf, incorporating all regression and classification residuals initially computed from the predicted default values $\theta_i^{(0)}$. It then appends branches to the tree by finding the best split that maximizes the overall gain G . Expanding w_{reg} and w_{cls} in (13) yields:

$$\tilde{L}^{(t)} = -\frac{1}{2} \sum_{j=1}^T \left[\frac{\left[\sum_{i \in I_j} (g_{reg_i}) \right]^2}{\sum_{i \in I_j} (h_{reg_i}) + \lambda} + \frac{\left[\sum_{i \in I_j} (g_{cls_i}) \right]^2}{\sum_{i \in I_j} (h_{cls_i}) + \lambda} \right] \quad (19)$$

where T denotes the number of leaves in the tree.

Equation (19) can be used as a scoring function to assess the quality of the tree structure. More specifically, a regression similarity score $S_{reg_{I_j}}$ and a classification similarity score $S_{cls_{I_j}}$ are calculated for each leaf j in the tree using:

$$S_{reg_{I_j}} = \frac{\left[\sum_{i \in I_j} (g_{reg_i}) \right]^2}{\sum_{i \in I_j} (h_{reg_i}) + \lambda} \quad (20)$$

$$S_{cls_{I_j}} = \frac{\left[\sum_{i \in I_j} (g_{cls_i}) \right]^2}{\sum_{i \in I_j} (h_{cls_i}) + \lambda} \quad (21)$$

To find the best split that maximizes the overall gain G , assume that I_L and I_R are the sets of instances in the left and right nodes after the split, where $I = I_L \cup I_R$. Using the similarity scores computed in (20) and (21), the regression gain G_{reg} and the classification gain G_{cls} are defined as:

$$G_{reg} = S_{reg_{I_L}} + S_{reg_{I_R}} - S_{reg_I} \quad (22)$$

$$G_{cls} = S_{cls_{I_L}} + S_{cls_{I_R}} - S_{cls_I} \quad (23)$$

where S_{reg_I} and S_{cls_I} denote the regression and classification similarity scores of the parent node before the split, respectively, while $S_{reg_{I_L}}$, $S_{reg_{I_R}}$, $S_{cls_{I_L}}$, and $S_{cls_{I_R}}$ represent the corresponding similarity scores for the left and right child nodes after the split. Hence, the overall gain G is obtained using:

$$G = G_{reg} + G_{cls} \quad (24)$$

where the regression gain G_{reg} and the classification gain G_{cls} are computed as shown in (22) and (23). The best splitting threshold that maximizes the overall gain, obtained from (24), is then selected, and a branch is added to the tree. The proposed split finding approach is outlined in Algorithm 2.

Building upon the described methodology and learning mechanism, the proposed DXGBA can be highly beneficial in numerous real-world applications where both classification and regression tasks occur concurrently. For example, in environmental monitoring, systems may simultaneously estimate the particulate concentration as a regression task and classify air quality as either safe or hazardous as a classification task. In the context of intelligent transportation, DXGBA would be

beneficial to simultaneously determine the estimated time of arrival (ETA) and classify road traffic conditions as congested or clear. In sports analytics, one may classify the outcome of a play as a win or a loss; and assess a player's contribution to the anticipated score or fatigue level. In predictive maintenance, the proposed approach would assist in simultaneously predicting the remaining useful life (RUL) and classifying the operational status of a machine as functional or faulty. Furthermore, intelligent tutoring systems in education technology have the potential to predict continuous learning performance scores or anticipated grades and classify the engagement levels of the students. Moreover, using DXGBA in customer service automation would simultaneously forecast the estimated resolution time based on the complexity of incoming queries and classify them as urgent or non-urgent. These examples show the potential of DXGBA to manage various and uncorrelated dual-output tasks.

Algorithm 2 Split Finding Algorithm

Input: j : leaf node to split I_j : instance set at node j with M features**Output:**

Best tree split with maximum gain

Initialize max gain: $G_{\max} \leftarrow 0$ Calculate regression and classification scores for I_j using (20) and (21).**for** each feature $m = 1$ to M **do**Sort I_j by feature m **for** each split point $n = 1$ to $|I_j|$ **do**1. Compute threshold τ_{nm} 2. Split I_j into I_L and I_R using τ_{nm} 3. Calculate regression and classification scores for I_L and I_R using (20) and (21).

4. Compute:

Regression gain $G_{reg_{nm}}$ using (22).Classification gain $G_{cls_{nm}}$ using (23).Combined gain G_{nm} using (24).5. Update $G_{\max} \leftarrow \max(G_{\max}, G_{nm})$ 6. Record the split point if $G_{nm} > G_{\max}$ **end for****end for****if** $G_{\max} > 0$ **and** max tree depth not reached **then**Split node j into j_{Left} and j_{Right}

Recurse:

call *splitFinding*(j_{Left} , I_L)**call** *splitFinding*(j_{Right} , I_R)

IV. EXPERIMENTS

In this research experiment, the proposed approach is evaluated using five different datasets. Namely, bankruptcy [34], diabetes [35], apple quality [36], credit card default [37], and bank marketing [38] datasets were used in this study. In particular, the bankruptcy prediction [34] dataset was employed to train the proposed model and predict company bankruptcy

based on relevant financial factors. It includes 6819 samples collected from the Taiwan Economic Journal between 1999 and 2009 [34]. This dataset encloses 94 features that represent relevant financial ratios and performance metrics. It is typically used to predict the gross profit-to-total-sales ratio as a regression target and to classify companies as either bankrupt or not. On the other hand, the diabetes dataset, collected from the National Institute of Diabetes and Digestive and Kidney Diseases [35], includes the health diagnostics and demographic data of 768 patients. Specifically, this dataset consists of 7 features, including glucose levels, blood pressure, and age, which represent various medical and personal attributes. Besides, the Body Mass Index (BMI) represents the regression target, whereas diabetic or nondiabetic represents the predefined classes. The apple quality dataset [36] is typically used to evaluate the apple quality based on relevant physical and chemical characteristics. It includes 3991 samples collected from the American Agriculture Company. The dataset contains 6 features, including size, weight, and sweetness. Moreover, the acidity level is used as a regression target. Besides, the class values reflect the apples' quality; acceptable or unacceptable. The default of the credit card clients dataset [37] was also associated with the proposed model to predict the likelihood of a credit card user defaulting on their payment based on related demographic features and financial behavior. It consists of 30,000 samples, obtained from credit card holders in Taiwan in April 2005. The dataset comprises 22 features, such as invoicing amount, payment history, and demographic information. It was employed to classify clients as either default or nondefault and estimate the credit limit as a regression target. Finally, the bank marketing dataset [38] was considered in this research to predict the effectiveness of a direct marketing campaign for a Portuguese bank based on customer demographics and campaign factors. It encloses 45,211 samples, obtained from the bank's marketing efforts throughout many years. The dataset includes 15 features, such as age, job type, and previous campaign outcomes. It was used to predict the balance amount as a regression target and classify whether a customer will subscribe to a term deposit or not. Table 2 overviews the considered datasets and shows the corresponding class distributions. As one can notice, all datasets are imbalanced except the apple quality.

The selection of benchmark datasets was motivated by the need to validate DXGBA across different application domains and problem structures. As can be noticed, the considered datasets span a wide range of real-world contexts. This will provide a thorough assessment of the proposed model and demonstrate the robustness and generalization capabilities in various data conditions. Moreover, to comply with problem requirements and the dual-task nature of the proposed approach, each data set includes a binary classification target and a continuous regression target. As mentioned earlier, the proposed approach is intended to address real-world problems where classification and regression are two distinctive tasks. Accordingly, these datasets are selected to ensure that the target variables are

uncorrelated. More specifically, the bankruptcy [34] dataset was selected to test the model on high-dimensional financial data with imbalanced classes. The diabetes [35] dataset represents a healthcare case with moderate feature dimensionality, which offers a real-world case where classification (diabetic status) and regression (BMI) are independent. The apple quality [36] dataset is a balanced dataset that involves physical product evaluation. The credit card default [37] and bank marketing [38] datasets were included to evaluate model performance in prediction tasks under varied class distributions and customer profiling features.

TABLE 2. Overview of the datasets used in this research.

Dataset	No. of Samples	No. of Features	Pos. Samples	Neg. Samples
Bankruptcy [34]	6819	94	220	6599
Diabetes [35]	768	7	268	500
Apple Quality [36]	3991	6	1998	1993
Credit Card [37]	30,000	22	6636	23,364
Bank [38]	45,211	15	5289	39,922

A. CONFIGURATION SETUP

All datasets were complete and contained no missing values, hence, no imputation or row removal was required. Each dataset was divided into 60% training, 20% validation, and 20% test sets without stratification. The classification performances of the considered models were assessed using the Accuracy, Precision, Recall, F1-score [39], and Area Under the Curve (AUC) [40] metrics. On the other hand, the regression performances of the models were evaluated using the Mean Absolute Error (MAE) [41] and the Mean Squared Error (MSE) [41]. Moreover, a 5-fold cross-validation was deployed to validate and assess all models, and all reported performance metrics were averaged across the five cross-validation folds. Neither repeated nor nested cross-validation was applied. Additionally, a feature scaling was conducted using Min-Max Normalization, and the categorical features were label encoded. To avoid overfitting, the training of each model was performed with early stopping and a patience of 5 rounds, monitored using the evaluation set. Besides, the proposed dual task approach was implemented using the Python programming language and the following libraries: XGBoost version 2.1.4, scikit-learn, pandas, and imbalanced-learn. The experiments were conducted using a Jupyter Notebook along with a local machine with 8 GB of RAM and an 8-core Apple M3 CPU. GPU acceleration was not utilized during experimentation.

Note that the model hyperparameters were tuned manually using the validation set. In particular, for the learning rate, various values were investigated. Specifically, 0.001, 0.003, 0.01, 0.03, and 0.1 were explored, and the learning rate was accordingly set to 0.1. Moreover, various values were experimented with for the maximum tree depth: 8, 16, 32, and 64. Consequently, the maximum depth was set to 16. Furthermore, the subsampling rates are varied and set

respectively to 0.6, 0.8, and 1. Accordingly, a sampling rate of 0.8 was selected. In addition, values for the L1 regularization term (α) were fixed at 0, while L2 regularization (λ) was explored in the range 0, 0.5, 1.3, and 10, with 1.3 selected. The gamma parameter, which controls minimum loss reduction for further tree splitting, was tested at 0, 0.5, 2, and 5, and a value of 2 was chosen. The initial prediction scores were set to 0.5 for all models. To ensure reproducibility, a fixed random seed of 42 was used across all random operations, including KFold splitting and train-test splitting. The hyperparameters employed in the proposed DXGBA approach are detailed in Table 3. It is important to note that the XGBoost models for both classification and regression tasks utilize identical hyperparameters, except for the objective function. Since DXGBA employs a customized objective function, the classification model depends on a “binary:logistic” objective function, while the regression model employs “reg:squarederror”.

TABLE 3. Hyperparameters used for the proposed DXGBA approach.

Hyperparameter	Value
learning rate	0.1
maximum tree depth	16
subsample ratio	0.8
colsample by tree	0.8
colsample by level	0.8
regularization term (λ)	1.3
tree pruning penalty (γ)	2
early stopping rounds	5
classification base score	0.5
regression base score	0.5
booster	gbtree

B. COMPARISON OF THE PROPOSED DXGBA WITH CONVENTIONAL MODELS

First, the performance of XGBoost [11] models is evaluated in addressing both classification and regression tasks. The goal of this experiment was to compare XGBoost’s performance with that achieved using state-of-the-art machine learning techniques typically used for regression and classification tasks. Namely, linear regression, logistic regression, Support Vector Machines [42] (SVMs), and Neural Networks [43] (NNs) were considered in this experiment. Table 4 summarizes the hyperparameters used for the models. For neural networks (NN), binary cross-entropy loss with sigmoid activation was applied to classification tasks, while mean squared error loss with linear activation was used for regression tasks.

As can be seen in Table TABLE 4, XGBoost demonstrated consistently a superior or competitive performance compared to other models. Particularly, it outperformed the other models based on metrics such as F1-score and AUC. For instance, for the apple quality dataset [36], XGBoost achieved the highest accuracy and F1-score, attaining 85.24% and 0.8544, respectively. Although Logistic Regression and SVM achieved competitive accuracy using the bankruptcy

dataset [34], the reported F1-score and recall were notably lower compared to the results achieved using XGBoost and NN. This can be attributed to their limitations in capturing nuanced patterns in the data. One should also note that Neural Networks performed overall well. However, they lag slightly behind XGBoost based on key metrics attainments. This showcases XGBoost's robustness and adaptability across diverse domains.

TABLE 4. Hyperparameters of conventional baseline models (Linear Regression, Logistic Regression, Support Vector Machine, and Neural Network).

Model	Hyperparameter	Value
Liner Regression	fit_intercept	True
	copy_X	True
	n_jobs	None
	positive	False
Logistic Regression	penalty	l2
	C	1.0
	solver	lbfgs
	fit_intercept	True
	max_iter	100
	class_weight	None
	random_state	42
	multi_class	auto
SVM	kernel	rbf
	C	1
	epsilon	0.1
	gamma	Scale
	shrinking	True
	tol	1e-3
	cache_size	200 (MB)
	max_iter	-1
NN	first hidden layer	64 units, ReLU
	second hidden layer	32 units, ReLU
	output Layer	1 unit, linear activation
	optimizer	Adam
	learning rate	0.001
	batch size	32
	epochs	50
	beta_1	0.9
	beta_2	0.999
	epsilon	1e-7

Table 6 details the results of the regression task obtained using the considered conventional models as well as XGBoost across all datasets. XGBoost demonstrated a superior or highly competitive performance. It particularly excelled in minimizing MAE and MSE using the bankruptcy and apple quality datasets. Specifically, for the bankruptcy dataset [34], it yielded the lowest MAE and MSE of 0.0094 and 0.0003, respectively. On the other hand, the Linear Regression performed well on simpler datasets like bank marketing [38]. However, it yielded significantly higher errors for more complex datasets, such as the bankruptcy dataset [34], with MAE and MSE much higher than those attained by the other models. In particular, NN and SVM yielded competitive performance using some datasets, but they are often trailed behind XGBoost in terms of precision. Overall, XGBoost's ability to deliver low error rates across diverse datasets demonstrates its robustness and effectiveness for the considered regression tasks.

Overall, XGBoost achieved consistently competitive or best performance in both classification and regression tasks. It often outperformed logistic regression, SVM, and NN and achieved the best Recall, F1-score, and AUC across all datasets, except the diabetes dataset, where XGBoost's AUC ranked slightly behind NN's AUC. Additionally, XGboost achieved the highest Recall, which makes it suitable for tasks where identifying the positive cases is essential. Moreover, it achieved a significant improvement in terms of F1-score, especially for the bankruptcy [34] and bank marketing [38] datasets. This proves its robustness and effectiveness in handling imbalanced datasets. It also yielded a better balance

between Precision and Recall for most datasets. One can notice that NN proved to be competitive in some cases and yielded near-best or best results. However, XGBoost overtook the other models in terms of Recall and F1-score using all datasets. Unlike logistic regression and SVM, XGBoost maintained consistency in performance for both classification and regression. For the regression tasks, XGBoost yielded the lowest MAE and MSE.

XGBoost's performance can be attributed to several factors. First, XGBoost encloses an ensemble learning strategy that facilitates the modeling of nonlinear and intricate relationships through the construction of a decision tree ensemble and the aggregation of their predictions. Conversely, the single learners fail to represent such complex relationships. Further consideration is the gradient boosting framework utilized by XGBoost. More specifically, XGBoost optimizes the objective function by iteratively minimizing the residuals of the prior predictions. This yields a better convergence compared to the other models that maximize a single loss. XGBoost ultimately offers feature importance metrics, including gain and cover. This enables it to leverage the features that achieve a significant impact on the intended task predictions.

C. COMPARISON OF THE PROPOSED DXGBA WITH RELEVANT CLASSIFICATION AND REGRESSION APPROACHES

Unlike traditional single-task approaches which the models are trained separately for the intended prediction task, classification or regression, the proposed approach is designed to predict both a continuous value and a binary class label within a single framework. Accordingly, in this experiment, the performance of the proposed simultaneous classification and regression model, DXGBA, is compared with those achieved using typical XGBoost models trained independently for regression and classification tasks. In particular, the results outlined in Table 7 demonstrate that the DXGBA model consistently overtakes the XGBoost model across both classification and regression tasks. As can be seen, in terms of classification performance, DXGBA achieves higher accuracy, precision, recall, F1-score, and AUC using all datasets, with notable improvements for recall and F1-score. This indicates better handling of class imbalances and overall classification performance. Consistently, DXGBA

TABLE 5. Classification results obtained using conventional models on five datasets (bold values represent the highest performance).

Dataset	Model	Accuracy	Precision	Recall	F1-Score	AUC
Bankruptcy [34]	Logistic Regression	96.95	0.7838	0.0911	0.1587	0.9177
	Neural Network	96.61	0.4971	0.1789	0.2367	0.903
	SVM	96.8	0.4	0.0168	0.032	0.8808
	XGBoost	96.79	0.5063	0.3247	0.3942	0.9382
Diabetes [35]	Logistic Regression	74.48	0.7063	0.4639	0.5559	0.8061
	Neural Network	75.39	0.6815	0.5581	0.6099	0.815
	SVM	73.96	0.6885	0.4696	0.5533	0.8011
	XGBoost	75.52	0.6811	0.5715	0.6179	0.8048
Apple Quality [36]	Logistic Regression	72.39	0.7149	0.7467	0.7303	0.796
	Neural Network	83.41	0.8359	0.8321	0.8336	0.9227
	SVM	73.26	0.7118	0.7836	0.7458	0.7976
	XGBoost	85.24	0.8435	0.8659	0.8544	0.9321
Default of Credit Card Clients [37]	Logistic Regression	80.83	0.7196	0.2201	0.3367	0.7138
	Neural Network	81.95	0.6745	0.359	0.4677	0.763
	SVM	81.65	0.6952	0.3052	0.4237	0.7076
	XGBoost	81.9	0.6684	0.3623	0.4695	0.7718
Bank Marketing [38]	Logistic Regression	89.03	0.605	0.1792	0.2764	0.865
	Neural Network	90.07	0.6279	0.3706	0.4631	0.9031
	SVM	89.12	0.6609	0.1455	0.2381	0.8767
	XGBoost	90.54	0.6342	0.4543	0.5291	0.9331

TABLE 6. Regression results obtained using conventional models on five datasets (bold values represent the highest performance).

Dataset	Model	MAE	MSE
Bankruptcy [34]	Logistic Regression	18.83	2.3×10^6
	Neural Network	0.01	0.0003
	SVM	0.0453	0.0025
	XGBoost	0.0094	0.0003
Diabetes [35]	Logistic Regression	0.0776	0.011
	Neural Network	0.0771	0.011
	SVM	0.0777	0.0111
	XGBoost	0.087	0.0138
Apple Quality [36]	Logistic Regression	0.1065	0.0183
	Neural Network	0.1106	0.0197
	SVM	0.1063	0.0184
	XGBoost	0.0929	0.0149
Credit Card [37]	Logistic Regression	0.0808	0.0111
	Neural Network	0.0973	0.0158
	SVM	0.0827	0.0112
	XGBoost	0.0816	0.0112
Bank [38]	Logistic Regression	0.0137	0.0007
	Neural Network	0.0161	0.0009
	SVM	0.0814	0.007
	XGBoost	0.0141	0.0008

yielded lower MAE and MSE values for the regression task, compared to XGBoost. These results suggest that the proposed simultaneous classification and regression approach, DXGBA, enhances the performance for both intended predictive tasks while maintaining computational efficiency. Specifically, the bankruptcy dataset [34] yielded the most significant increase in accuracy, with 97.05% and 96.79% for the proposed DXGBA and XGBoost, respectively. Furthermore, this dataset resulted in a noticeable improvement in terms of F1-score. DXGBA attained an F1-score of 0.4718, whereas XGBoost's F1-score has not exceeded 0.3942. Although the improvements are less pronounced for the other datasets, DXGBA consistently outperformed XGBoost, highlighting its effectiveness in the intended classification tasks. Regarding regression performance, DXGBA surpassed XGBoost

with lower MSE values across all datasets. The most notable improvement is observed for the diabetes dataset [35]. DXGBA achieved an MSE of 0.011, whereas XGBoost's MSE was 0.0138.

These results demonstrate the ability of the proposed DXGBA's dual learning approach in leveraging the mutual benefits of both classification and regression techniques. In other words, DXGBA optimally utilizes features by evaluating their impacts on both tasks concurrently. For example, the features that exhibit weak correlation with the regression target may be significant for the classification task. This is accomplished by optimizing a cost function that integrates gradients and Hessians reflecting both tasks.

Moreover, Table 8 presents the training and testing times recorded in seconds for each model. One should note that XGBoost was trained and tested separately for each task. The reported execution time represents the resulting sum. As one can see, DXGBA exhibits an overall time efficiency, especially in testing. This makes it more suitable for real-time applications. As can also be observed, DXGBA yielded lower testing time compared to XGBoost using all datasets. In addition, DXGBA showcased faster training time for all datasets, except the default of the credit card clients dataset [37]. This can be attributed to the latter dataset's complexity. However, as the training is typically conducted offline, the time has a minor impact on real-time applications. For the training time, the smallest decrease, from 9.9s to 7.42s, was recorded for the apple quality dataset [36]. On the other hand, the bankruptcy dataset [34] yielded a drastic decrease in the training time from 7.3s for XGBoost to 4.24s for DXGBA. These results demonstrate DXGBA's capability to reduce redundant computations when achieving simultaneous classification and regression. In particular, the testing time reduction for DXGBA compared to XGBoost is approximately 81.9% for bankruptcy [34], 48% for diabetes [35],

TABLE 7. Results achieved by DXGBA and XGBoost on the five datasets (bold values represent the highest performance).

Dataset	Model	Classification				Regression		
		Accuracy	Precision	Recall	F1-Score	AUC	MAE	MSE
Bankruptcy [34]	XGBoost	96.79	0.5063	0.3247	0.3942	0.9382	0.0094	0.0003
	DXGBA	97.05	0.5626	0.414	0.4718	0.9402	0.0078	0.0002
Diabetes [35]	XGBoost	75.52	0.6811	0.5715	0.6179	0.8048	0.087	0.0138
	DXGBA	75.91	0.6823	0.5865	0.6276	0.8136	0.0756	0.011
Apple Quality [36]	XGBoost	85.24	0.8435	0.8659	0.8544	0.9321	0.0929	0.0149
	DXGBA	85.52	0.8455	0.8693	0.8571	0.9332	0.0899	0.0136
Default of Credit Card Clients [37]	XGBoost	81.9	0.6684	0.3623	0.4695	0.7718	0.0816	0.0111
	DXGBA	81.99	0.6697	0.368	0.4748	0.7728	0.0736	0.01
Bank Marketing [38]	XGBoost	90.54	0.6342	0.4543	0.5291	0.9331	0.0141	0.0008
	DXGBA	90.62	0.6363	0.4622	0.5353	0.9337	0.0135	0.0007

TABLE 8. Training and testing times achieved by DXGBA and XGBoost Models (bold values indicate the shortest times recorded).

Dataset	Model	Training Time (s)	Testing Time (s)
Bankruptcy [34]	XGBoost	7.3008	0.0215
	DXGBA	4.2358	0.0039
Diabetes [35]	XGBoost	0.5268	0.0025
	DXGBA	0.3069	0.0013
Apple Quality [36]	XGBoost	9.9028	0.0467
	DXGBA	7.4219	0.0109
Credit Card [37]	XGBoost	31.6868	0.5111
	DXGBA	72.046	0.4998
Bank [38]	XGBoost	7.0199	0.0633
	DXGBA	4.2542	0.0281

76.7% for apple quality [36], 2.2% for default of credit card clients [37], and 55.6% for bank marketing [38].

Moreover, the results obtained using the proposed DXGBA are compared with the best-performing classification and regression models. Initially, the proposed DXGBA model, which handles both regression and classification tasks simultaneously, is trained on the considered datasets. The results are then compared with those obtained using the best-performing models in each task separately. As can be seen in Table 9, in terms of classification performance, DXGBA consistently outperforms the individual best-performing models across all datasets. In particular, for the bankruptcy dataset [34], DXGBA achieved the highest improvement in terms of F1-score with scores of 0.4718 and 0.3942, showcasing its ability to handle imbalanced classes more effectively. Similarly, for the diabetes dataset [35], DXGBA yielded a higher F1-score of 0.6276 compared to XGBoost, indicating its robustness in achieving a better trade-off between precision and recall. For the other datasets, such as apple quality [36], default of credit card clients [37], and bank marketing [38], DXGBA also yielded moderate, but consistent, improvements in terms of Precision, Recall, and AUC. Similarly, for the regression tasks, DXGBA demonstrates superior performance by achieving lower MAE and MSE across all datasets.

This performance may be attributed to the simultaneous learning conducted by the proposed DXGBA, which enables the model to share information between the classification and

the regression tasks. Unlike XGBoost, the proposed DXGBA identifies the prevalent patterns in the data that are relevant to both tasks. Consequently, using such shared information enhances both tasks' performance. The most significant improvement was recorded for the bankruptcy dataset [34]. Specifically, DXGBA reduced the MSE from 0.0003 to 0.0002, indicating a better predictive precision for continuous variables. For the diabetes dataset [35], DXGBA attained the same MSE achieved by the NN model at 0.011. This showcases its competitiveness with the considered single-task models. Additionally, for the apple quality dataset [36], DXGBA improved in terms of MSE with scores of 0.0136 and 0.0149, reinforcing its ability to handle regression tasks with better accuracy. These results underline the advantage of the proposed DXGBA's unified approach, which leverages joint learning for more accurate predictions.

D. COMPARISON OF THE PROPOSED DXGBA WITH STATE-OF-THE-ART APPROACHES

In the subsequent experiment, the performances achieved using the proposed approach are compared with those obtained using the state-of-the-art approaches that perform simultaneous classification and regression. One should note that the considered relevant works exhibit different architectures and techniques. Namely, the state-of-the-art approaches are: DualTaskNet [44], Joint RRN Framework [17] and Aux-LSTM [19]. As mentioned previously, DualTaskNet [44] employs a neural network architecture with two output layers to perform simultaneous classification and regression tasks through dual learning. Alternatively, the Joint RRN Framework [17] approach uses a joint classification-regression Recurrent Neural Network [45] (RNN). The network is trained end-to-end by optimizing a joint objective function. Also, Aux-LSTM [19] combines classification and regression models in a single framework using LSTM layers, which are jointly learned. These studies are directly comparable to DXGBA's dual-task approach and align well with its capability of simultaneous learning. Table 10 lists the hyperparameters utilized for the state-of-the-art approaches evaluated in this study. As can be seen in Table 11, DXGBA consistently outperforms the other models in both classification and regression tasks. Moreover, it achieved the highest

TABLE 9. Performance measures achieved by the proposed DXGBA and the best-performing single-task models across five datasets (bold values represent the highest performance).

Dataset	Model(s)	Classification				Regression		
		Accuracy	Precision	Recall	F1-Score	AUC	MAE	MSE
Bankruptcy [34]	XGBoost	96.79	0.5063	0.3247	0.3942	0.9382	0.0094	0.0003
	DXGBA	97.05	0.5626	0.414	0.4718	0.9402	0.0078	0.0002
Diabetes [35]	XGBoost / NN	75.52	0.6811	0.5715	0.6179	0.8048	0.0771	0.011
	DXGBA	75.91	0.6823	0.5865	0.6276	0.8136	0.0756	0.011
Apple Quality [36]	XGBoost	85.24	0.8435	0.8659	0.8544	0.9321	0.0929	0.0149
	DXGBA	85.52	0.8455	0.8693	0.8571	0.9332	0.0899	0.0136
Default of Credit Card Clients [37]	XGBoost / Linear Regression	81.9	0.6684	0.3623	0.4695	0.7718	0.0808	0.0111
	DXGBA	81.99	0.6697	0.368	0.4748	0.7728	0.0736	0.01
Bank Marketing [38]	XGBoost / Linear Regression	90.54	0.6342	0.4543	0.5291	0.9331	0.0137	0.0007
	DXGBA	90.62	0.6363	0.4622	0.5353	0.9337	0.0135	0.0007

accuracy, F1-score, and AUC while maintaining the lowest regression errors for all datasets.

Furthermore, DXGBA demonstrated a significant improvement in accuracy, particularly for the diabetes [35] and apple quality [36] datasets. Besides, the gains in terms of F1-score and AUC are also substantial, for the bankruptcy [34] and apple quality [36] datasets. The highest increases in F1-score and AUC attained 0.17 and 0.08, respectively. As one can also observe, Aux-LSTM [19] has fallen consistently behind the other models. For the Precision metric, DXGBA demonstrated a considerable improvement for the bankruptcy [34] and apple quality [36] datasets. However, the gain was marginal for the default of credit card clients [37] dataset. Evidently, Joint RNN Framework [17] and DualTaskNet [44] yielded a slightly higher Precision than DXGBA for the bankruptcy [34] and bank marketing [38] datasets. However, as the datasets are imbalanced, these improvements in Precision lead to a decrease in Recall. Similarly, Aux-LSTM [19] achieved the highest Recall for the bankruptcy [34] and bank marketing [38] datasets, resulting in a significant reduction in Precision. Besides, DXGBA excelled in improving Recall. For the regression, DXGBA achieved the lowest MAE and MSE using all datasets. This confirms its efficiency and robustness in achieving the intended prediction tasks. The remarkable success of DXGBA compared to contemporary approaches may be attributed to the fact that those challenger models exhibit low generalization ability. In other words, they may occasionally excel on a specific dataset but struggle to maintain a consistent performance across various datasets. Moreover, DXGBA's architecture associates the gradient boosting and the ensemble learning approaches to adapt to varying data distributions and tasks, in contrast to the sequential and recurrent architectures of other models. Furthermore, DXGBA equilibrates various metrics, whereas competing models aim at optimizing a single metric.

To determine whether the differences in performance achieved by the proposed DXGBA and the state-of-the-art models are statistically significant, the results must be statistically validated. Parametric tests are the preferred method for validating results. Nevertheless, it is essential to initially prove the safe use of these procedures. Independence,

normality, and homoscedasticity are the three criteria that must be satisfied [46]. The independence of observations was satisfied, as each fold represented a separate, non-overlapping subset of the data, and model evaluations were performed independently per fold. Notably, the paired t-test does not require the assumption of homoscedasticity as it focuses on the differences within pairs rather than comparing variances across groups [47]. To evaluate the normality condition, the Shapiro–Wilk test [48] was conducted. Since the resulting p-values were greater than 0.05 in most cases, the normality assumption was satisfied for the majority of model comparisons. However, five specific cases exhibited p-values below 0.05, indicating deviations from a normal distribution. As a result, parametric tests were inapplicable for those comparisons. For model comparisons where the normality assumption was satisfied, the parametric paired t-test [49] was applied. In cases where normality was not met, the non-parametric Wilcoxon signed-rank test [50] was used instead. Table 12 provides a comprehensive summary of the obtained test results. Specifically, the p-values obtained using DXGBA along with DualTaskNet [44], Joint RNN Framework [17], and Aux-LSTM [19] across all datasets are reported. It is also important to note that the reported p-values were obtained using the F1-score and MSE results.

As it can be seen, DXGBA yielded statistically significant enhancements in both classification and regression tasks. This finding holds for the majority of the considered datasets. More specifically, DXGBA significantly outperformed other models on most F1-score and MSE comparisons, as confirmed by paired t-tests when normality was satisfied. Nevertheless, a small number of comparisons did not demonstrate statistically significant differences. These include the classification performance on the diabetes [35] dataset when compared to DualTaskNet [44], and the regression performance against Aux-LSTM [19]. Similarly, in the apple quality [36] dataset, DXGBA did not substantially outperform the Joint RNN Framework [17] and Aux-LSTM [19] in F1-score. This may be attributed to the violation of normality assumptions and the conservative nature of the Wilcoxon test [51]. In the default of credit

TABLE 10. Hyperparameters used for the state-of-the-art approaches.

Model	Hyperparameter	Value
DualTaskNet [44]	first dense layer	64 units, ReLU activation
	second dense layer	32 units, ReLU activation
	third dense layer	1 unit, sigmoid activation
	classification output layer	binary_crossentropy
	regression output layer	MSE
	optimizer	Adam
	loss for classification	32
	loss for regression	50
	loss weights	0.9 (cls), 0.1 (reg)
	epochs	50
	batch size	32
	learning rate	0.001
Joint RNN Framework [17]	rho	0.9
	momentum	0
	epsilon	1e-7
	number of LSTM layers	3
	number of FC layers	3
	number of neurons per layers	100
	optimizer	SGD with momentum
	batch size	32
Aux-LSTM [19]	learning rate	0.01
	epochs	50
	classification activation	Softmax
	regression activation	ReLU
	LSTM hidden layer dimension	128
	fully connected layer dimension	64
	learning rate	0.01
	dropout probability for regression	0.25
	dropout probability for classification	0.5
	epochs	15
	loss combination weight (λ)	0.75

TABLE 11. Performance measures achieved by the proposed DXGBA and relevant state-of-the-art approaches across five datasets (bold values represent the highest performance).

Dataset	Model	Classification				Regression		
		Accuracy	Precision	Recall	F1-Score	AUC	MAE	MSE
Bankruptcy [34]	DualTaskNet [44]	96.8	0.4	0.0154	0.0293	0.9059	0.132	0.0198
	Joint RNN Framework [17]	96.69	0.4486	0.1473	0.2047	0.9005	0.0098	0.0003
	Aux-LSTM [19]	89.04	0.1967	0.7332	0.307	0.8945	0.0137	0.0004
	DXGBA	97.05	0.5626	0.414	0.4718	0.9402	0.0078	0.0002
Diabetes [35]	DualTaskNet [44]	69.4	0.601	0.4873	0.4937	0.7684	0.102	0.0174
	Joint RNN Framework [17]	72.01	0.689	0.4475	0.5144	0.7973	0.0882	0.014
	Aux-LSTM [19]	65.24	0.1333	0.0085	0.016	0.7227	0.0913	0.0151
	DXGBA	75.91	0.6823	0.5865	0.6276	0.8136	0.0756	0.011
Apple Quality [36]	DualTaskNet [44]	73.26	0.7012	0.8613	0.7635	0.8552	0.129	0.026
	Joint RNN Framework [17]	72.51	0.6774	0.8561	0.7562	0.7936	0.1082	0.0189
	Aux-LSTM [19]	57.36	0.488	0.6418	0.5273	0.732	0.1148	0.0206
	DXGBA	85.52	0.8455	0.8693	0.8571	0.9332	0.0899	0.0136
Default of Credit Card Clients [37]	DualTaskNet [44]	81.71	0.6666	0.3592	0.4631	0.7514	0.0767	0.0113
	Joint RNN Framework [17]	81.72	0.672	0.3495	0.4561	0.7573	0.0802	0.0109
	Aux-LSTM [19]	80.29	0.3884	0.2367	0.2938	0.6911	0.0757	0.0105
	DXGBA	81.99	0.6697	0.368	0.4748	0.7728	0.0736	0.01
Bank Marketing [38]	DualTaskNet [44]	89.79	0.6456	0.3203	0.4124	0.8943	0.0197	0.0011
	Joint RNN Framework [17]	89.95	0.6216	0.377	0.4654	0.8946	0.0138	0.0008
	Aux-LSTM [19]	79.69	0.3466	0.8231	0.4874	0.884	0.0159	0.0009
	DXGBA	90.62	0.6363	0.4622	0.5353	0.9337	0.0135	0.0007

card clients [37] dataset, no significant advances in F1-score were observed when compared to any of the three models. However, significant gains were observed in MSE across all comparisons. Lastly, DXGBA demonstrated statistically significant enhancements in both F1-score and MSE across all models in the bankruptcy [34] and bank marketing [38]

datasets, with the exception of regression performance in comparison to Aux-LSTM [19], which failed to satisfy normality.

To provide a comprehensive evaluation of the proposed DXGBA framework, DXGBA was compared with recent developments in metaheuristic hyperparameter optimization

TABLE 12. Statistical significance testing (paired t-test and Wilcoxon signed-rank test) of DXGBA compared to state-of-the-art models across five datasets for both classification (F1-score) and regression (MSE) tasks.

Dataset	Model	Metric	Normality Satisfied	p-value (Wilcoxon)	p-value (t-test)	Significant?
Bankruptcy [34]	DualTaskNet [44]	F1-score	Yes	-	0.0004	Yes
		MSE	Yes	-	0.025	Yes
	Joint RNN Framework [17]	F1-score	Yes	-	0.043	Yes
		MSE	Yes	-	0.029	Yes
	Aux-LSTM [19]	F1-score	Yes	-	0.005	Yes
		MSE	No	0.0625	NA	No
Diabetes [35]	DualTaskNet [44]	F1-score	Yes	-	0.187	No
		MSE	Yes	-	0.035	Yes
	Joint RNN Framework [17]	F1-score	Yes	-	0.014	Yes
		MSE	Yes	-	0.001	Yes
	Aux-LSTM [19]	F1-score	Yes	-	0.00001	Yes
		MSE	Yes	-	0.087	No
Apple Quality [36]	DualTaskNet [44]	F1-score	Yes	-	0.001	Yes
		MSE	Yes	-	0.00006	Yes
	Joint RNN Framework [17]	F1-score	No	0.0625	NA	No
		MSE	Yes	-	0.001	Yes
	Aux-LSTM [19]	F1-score	No	0.0625	NA	No
		MSE	Yes	-	0.001	Yes
Default of Credit Card Clients [37]	DualTaskNet [44]	F1-score	Yes	-	0.521	No
		MSE	Yes	-	0.001	Yes
	Joint RNN Framework [17]	F1-score	Yes	-	0.32	No
		MSE	Yes	-	0.002	Yes
	Aux-LSTM [19]	F1-score	No	0.8125	NA	No
		MSE	Yes	-	0.033	Yes
Bank Marketing [38]	DualTaskNet [44]	F1-score	Yes	-	0.033	Yes
		MSE	Yes	-	0.023	Yes
	Joint RNN Framework [17]	F1-score	Yes	-	0.003	Yes
		MSE	Yes	-	0.006	Yes
	Aux-LSTM [19]	F1-score	Yes	-	0.001	Yes
		MSE	No	0.0625	NA	No

TABLE 13. Comparison of classification and regression results of the proposed DXGBA model using metaheuristic optimization (MSCHO-DXGBA and manually tuned DXGBA across five datasets (bold values represent the highest performance).

Dataset	Model	Classification				Regression		
		Accuracy	Precision	Recall	F1-Score	AUC	MAE	MSE
Bankruptcy [34]	MSCHO- DXGBA	96.96	0.5562	0.2580	0.3476	0.9411	0.0067	0.0002
	DXGBA	97.05	0.5626	0.414	0.4718	0.9402	0.0078	0.0002
Diabetes [35]	MSCHO- DXGBA	72.52	0.6308	0.5122	0.5590	0.7835	0.0710	0.0101
	DXGBA	75.91	0.6823	0.5865	0.6276	0.8136	0.0756	0.0110
Apple Quality [36]	MSCHO- DXGBA	85.04	0.8451	0.8581	0.8514	0.9300	0.0924	0.0141
	DXGBA	85.52	0.8455	0.8693	0.8571	0.9332	0.0899	0.0136
Default of Credit Card Clients [37]	MSCHO- DXGBA	81.97	0.6760	0.3568	0.4664	0.7718	0.0737	0.0099
	DXGBA	81.99	0.6697	0.368	0.4748	0.7728	0.0736	0.01
Bank Marketing [38]	MSCHO- DXGBA	90.62	0.6369	0.4619	0.5353	0.9312	0.01357	0.00074
	DXGBA	90.62	0.6363	0.4622	0.5353	0.9337	0.01348	0.00070

for Extreme Gradient Boosting models. The Modified Sinh-Cosh Optimizer (MSCHO), as presented in [52], was integrated into DXGBA to enable the model to dynamically adjust its configuration by exploring the parameter space more effectively. MSCHO is a Sinh-Cosh Optimizer [53] (SCHO) with two enhancements. Firstly, Quasi-Reflective Learning (QRL) is incorporated to add diversity early by generating additional solutions. Secondly, inspired by the Artificial Bee Colony [54] (ABC) algorithm, the agent is replaced if it does not improve for two iterations.

In this experiment, six XGBoost hyperparameters were tuned—namely, learning rate, minimum child weight, subsample, colsample by tree, max depth, and gamma—using

the same value ranges reported in the original study. Table 13 reports the classification and regression results of DXGBA incorporating hyperparameter optimization compared to the manually tuned DXGBA. As can be seen, the two approaches performed comparably on both tasks across all datasets, with manually tuned DXGBA performing slightly better in most cases. More specifically, manually tuned DXGBA showed consistent classification performance, especially in recall and F1-score, which are critical for imbalanced or high-risk domains. Regarding regression, results were close in most cases, with minor improvements or equivalence in some cases. This experiment suggests that MSCHO-based tuning provides a competitive alternative, particularly when domain

knowledge or time for manual tuning is limited. So, while manually tuned DXGBA performed slightly better in this study and was faster to deploy, especially when sufficient expertise in parameter tuning is available, MSCHO-DXGBA is highly adaptable for unknown data characteristics as it explores a wider search space.

V. CONCLUSION

This paper presented DXGBA, a Dual-learning XGBoost Based Approach designed to perform classification and regression simultaneously using a shared tree structure and a joint objective function. By optimizing both tasks concurrently, DXGBA reduces redundant computation, improves efficiency, and captures shared patterns in the data. Empirical results across five real-world datasets from diverse application domains demonstrated that DXGBA consistently outperforms traditional single-task models and state-of-the-art methods in terms of F1-score, mean squared error, and testing time.

The proposed method has shown strong applicability in domains requiring dual-output modeling, such as intelligent transportation, predictive maintenance, and customer service automation. However, several limitations remain. The complexity of optimizing a joint objective function may reduce model interpretability compared to single-task counterparts, particularly in sensitive domains like healthcare or finance. The method also depends on careful hyperparameter tuning, lacks uncertainty estimation, and is sensitive to data imbalance, though resampling strategies can mitigate this issue.

A promising direction for future research is to integrate neural networks with the DXGBA framework to better model complex, non-linear relationships, especially in high-dimensional or multi-modal data. For instance, neural embeddings from image, text, or sequential inputs could be combined with structured features to enhance prediction. Such hybrid architectures may improve generalization, robustness, and accuracy by leveraging the strengths of both deep learning and gradient boosting.

Future work will also explore adaptive model structure optimization and evaluate DXGBA in domain-specific scenarios, particularly in healthcare and financial risk analysis, where dual-task learning and efficient, interpretable predictions are essential.

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